MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, HYDRAULIC, TYPE MJ-1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote-indicating, 320° movement, 4,000-psi, synchro-style, hydraulic pressure transmitter, designated Type MJ-1.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-C-5541	Chemical Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-A-8625 Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-8775 Hydraulic System Components, Aircraft and Missiles, General Specification for
MIL-B-27497 Bearing, Jewel, Sapphire or Ruby, Synthetic

STANDARDS

FEDERAL

* FED-STD-595 Colors

MILITARY

MIL-STD-100 Engineering Drawing Practices
* MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage
MIL-STD-130 Identification Marking of U.S. Military Property
MIL-STD-143 Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461 Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704 Aircraft Electric Power, Characteristics
MIL-STD-781 Reliability Tests Exponential Distribution
MIL-STD-810 Environmental Test Methods and Engineering Guidelines
MIL-STD-831 Test Reports, Preparation of
MIL-STD-889 Dissimilar Metals
MIL-STD-1186 Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS28005 Transmitter, Pressure, Synchro, Aircraft
MS33678 Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951-82 Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

* 3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials. Materials used in the manufacture of transmitters shall be suitable for the purpose and shall conform to applicable Government specifications. Materials conforming to contractor's specifications may be used provided the specifications are approved by the Government and contain provisions for adequate tests. The use of the contractor's specification will not constitute waiver of Government inspection.

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are to be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.3.2 Metals. Metals shall be corrosion-resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter, except where magnetic materials are essential.

3.3.4 Corrosive fumes. Materials used in the construction of the transmitter shall not produce corrosive or deleterious fumes under the test conditions specified herein.

3.3.5 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction.

3.4.1 Design. The transmitter shall be designed to operate with indicators conforming to MIL-I-7057.

3.4.1.1 Outline dimensions. The outline dimensions of the transmitter shall conform to MS28005-8.

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3.4.1.2 Adjustments. The transmitter shall be provided with a means for adjusting or correcting indications which shall be simple to manipulate and such that adjustments can be accomplished with tools ordinarily available. The means of adjustment shall be accessible from the exterior of the case and shall provide a minimum of 14 angular degrees of adjustment of the synchro rotor.

3.4.1.3 Mechanisms. Transmitter mechanisms shall be comprised of units distinct from their cases. The units shall be sufficiently rugged so that small distortions of the cases will not affect the accuracy of the indications.

3.4.1.4 Attitude. The transmitter shall be so designed that it will operate as specified herein in any position.

3.4.1.5 Gears. When gears are used in the transmitter, they shall be stainless steel with a minimum Rockwell hardness of 50C. An antibacklash gear system shall be used to eliminate errors. Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, or other irregularities. A minimum mesh of .031 inch shall be provided when gears are used.

3.4.1.6 Lubrication. The transmitter shall be so designed that no periodic lubrication will be required during service.

3.4.1.7 Reliability. The transmitter shall have a minimum acceptable meantime-between-failures (MTBF) of 2,000 hours based on a discrimination ratio of 1.5 and a confidence level of 90 percent.

3.4.1.8 Longevity. The transmitter shall have an operating life span (equipment longevity as defined in MIL-STD-781) of 3,000 hours during which the MTBF specified shall be demonstrated.

3.4.1.9 Overpressure stop. An overpressure stop shall be provided to restrain the prime mover. The stop shall permit the indicator pointer to travel beyond full scale to a point approximately midway between the last and zero graduations.

3.4.2 Construction. The transmitter shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall neither bind nor shake and shall be as nearly frictionless as possible.

3.5 Performance. The transmitter shall operate as specified herein under the following conditions:

a. Temperature: Operating temperatures ranging from -54° to +145°C and storage temperatures ranging from -62° to +160°C.

b. Vibration: Vibration of 10g from 5 to 2,000 Hz without vibration isolators

c. Temperature-altitude: Pressure equivalent to 100,000-foot altitude at -54°C

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d. Salt fog: Exposure to salt sea atmosphere for 48 hours

e. Shock of 20g for 11 milliseconds.

3.5.1 Tolerances. The transmitter shall operate within the tolerances specified in tables I and II.

TABLE I. Scale Error Tolerances

			Tolerance ± PSI	
Test Pressure (PSI)	Room Temp	Vibration Overpressure Seasoning	Low Temp (-54°C) High Temp (+70°C) Temperature Altitude	High Temp (145°C)
0	50	75	75	100
500	50	75	100	125
1,000	50	75	100	125
1,500	50	75	100	125
2,000	50	75	125	150
2,500	50	75	125	150
3,000	50	75	125	150
3,500	75	100	125	150
4,000	75	100	125	150

TABLE II. Friction, Position, and Pointer Error Tolerances

Test	Tolerance ± PSI
Friction	
Room temperature	50
High temperature (70°C)	50
High temperature (145°C)	75
Low temperature exposure	75
Position error at 2,000 psi	50
Leakage	25
Vibration	
Pointer variation	30
Pointer oscillation	60

3.5.2 Effect of lead length. When the lead length (No. 20 AWG stranded copper wire) between the transmitter and indicator synchro stators is changed from 5 to 300 feet, the indication shall not change more than 1° of arc.

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3.5.3 Leakage. Case leakage shall not exceed 1.0 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

3.5.4 Dielectric strength. The transmitter shall withstand application of 200V rms at a commercial frequency for a minimum period of 5 seconds.

3.5.5 Voltage and frequency variation. The transmitter shall withstand any combination of voltage and frequency variations from 22 to 30V and 320 to 480 Hz.

3.5.6 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 for class ID equipment.

3.5.7 Overpressure. The transmitter shall withstand an overpressure of 1,000 psi (5,000 psi applied pressure) for a 10-minute period.

3.5.8 Indicator operation. The transmitter shall be capable of operating either one or two indicators, which incorporate size 10 synchros, without a change in output equivalent to more than 1° of arc.

3.5.9 Burst pressure. The transmitter shall withstand a burst pressure of 7,500 psi.

3.5.10 Damping. When 4,000 psi is applied to the transmitter, the time required for the pressure to drop to 500 psi when pressure is removed shall be 1.0 second to 3.0 seconds.

3.5.11 Seasoning. The transmitter shall withstand 20,000 applications of pressure as required to produce indications from 0 to 85 $\pm$5 percent of full scale and return to 0 at a rate of 3 $\pm$1 cpm.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 Case. The case shall be made of metal and shall be hermetically sealed. The case shall be finished in durable dull black and shall be adequately protected to withstand the environmental conditions specified herein. The case shall incorporate a tear band so that the internal mechanism may be removed, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. Hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the transmitter may be subjected.

* 3.7.1 Filling medium. The case shall be filled with a suitable fluid to provide viscous damping of vibration and aircraft pump pulsations. Allowance shall be made for expansion of the damping fluid at high temperature by partially filling the case with helium gas. The helium gas shall contain not more than 0.006 milligrams of water vapor per liter (dew point -65°C) at the filling pressure. The absolute pressure of the helium in the case shall be 1 $\pm$0.1 atmosphere.

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3.7.2 Case pressure release. A metal blowout device shall be provided to discharge excessive pressure in the case. The device shall release the * internal pressure through a 0.062 inch nominal diameter orifice when the case pressure reaches 80 ±25 psi. The device shall not affect hermetic sealing or the results of the sealing test.

3.7.3 Hydraulic connection and case pressure release boss. The transmitter shall provide for straight thread hydraulic connection as indicated on MS28005-8. The case pressure release boss shall be located at the back of the transmitter case, in the general area of the vent boss shown on MS28005. No vent boss is required. The pressure boss and the case pressure release boss shall be identified with permanently inscribed P and CPR, respectively. The lettering shall be lusterless white, color No. 37038 of FED-STD-595, and shall be 0.030 inch high.

3.8 Electronic components. Electronic components and the application thereof shall be in accordance with MIL-E-5400. Nonstandard parts shall not be used except with prior approval of the procuring activity.

3.9 Electrical connector. A hermetically sealed MS33678R14S-2P electrical connector shall be provided.

* 3.10 Electrical characteristics. The transmitter synchro shall meet the requirements of 3.10.1 through 3.10.3 with a normal excitation of 26 V 400-Hz single-phase power applied to the primary (rotor).

3.10.1 Open secondary. The current drawn by the self-synchronous motor *shall not exceed 500 ma with secondaries (stator) open circuited.

* 3.10.2 Power input. Power consumption of the transmitter synchro shall not exceed 13.0 va.

3.10.3 Secondary voltage. The open-circuit voltage measured between any two secondary leads shall be 11.8 ±0.3V. The variation between the three maximum secondary voltages on any one synchro shall not exceed 0.2V.

3.11 Wiring

3.11.1 External. The electrical receptacle shall be so wired that the 26V, 400-Hz power supply will be connected to terminals A and B. External wiring between the transmitter and indicator shall be as shown on figure 1.

3.11.2 Internal. Internal wiring of the transmitter shall be color coded and shall be connected to the receptacle pins as shown on figure 2. When gears are used to drive the transmitter counterclockwise for increasing values of the parameter being measured, terminals S1 and S3 shall be reversed internally to cancel the effects of the gear reversal to assure clockwise rotation of the indicator synchro shaft and pointer with increasing values. Increasing pressure function input shall give ADC signal rotation at the connector and electrical zero shall be at 2,000 psi.

3.12 Jewel bearings. When used, synthetic sapphire or ruby jewel bearings shall be in accordance with MIL-B-27497.

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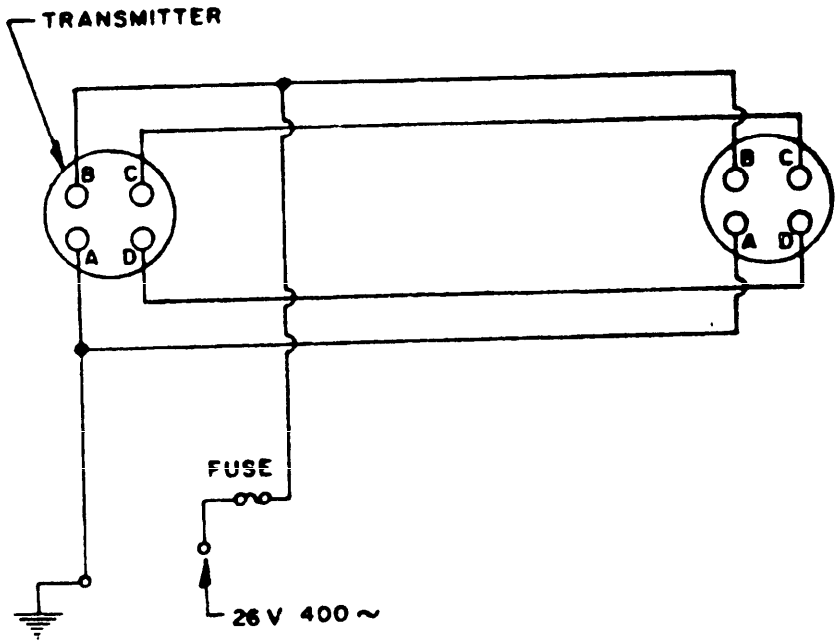
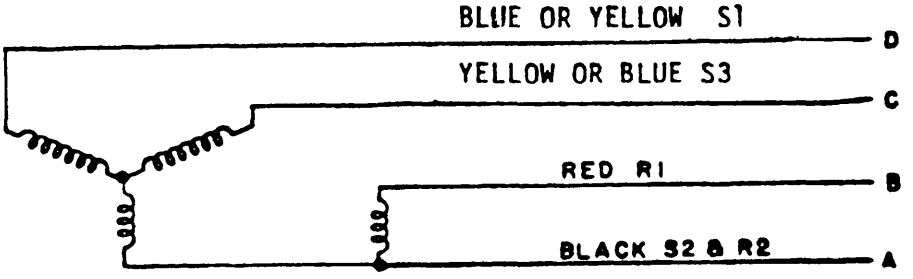


FIGURE 1. Wiring diagram - External, Single Indicator

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REQUIRED TO PROVIDE CLOCKWISE POINTER MOVEMENT ON THE INDICATOR

FIGURE 2. Internal Wiring - Transmitter

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3.13 Finishes and protective coatings

3.13.1 Aluminum alloy parts. Aluminum alloy parts shall be coated with an anodic film conforming to MIL-A-8625 except that small holes, straight threads, case inserts, and parts in hermetically sealed enclosures need not be anodized.

* 3.13.1.1 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film conforming to MIL-C-5541 plating or a corrosive resistant material that will not affect the function of the transmitter.

3.13.1.2 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.14 Weight. The weight of the transmitter shall not exceed 1.5 pounds.

3.15 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.16 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.16.1 Nameplate. A nameplate shall be permanently attached to the rear of the case. Where applicable, the information marked thereon shall conform to MIL-STD-130. The nameplate shall not be glued or cemented to the case nor shall metal decals be used.

3.16.2 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerance affect the interchangeability, operation, or performance of the transmitter, they shall be held or limited accordingly.

3.17 Workmanship. The transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

* a. First article inspection (see 4.4).

* b. Quality conformance inspection (see 4.5).

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4.3 Test conditions

4.3.1 Standard atmospheric conditions. Wherever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Tapping. Unless otherwise specified, the transmitter shall be lightly tapped or vibrated before each reading.

4.3.3 Test voltage. Unless otherwise specified, the transmitter shall be tested with 26V, 400-Hz, single-phase power. The voltage and frequency shall be maintained within ± 1 percent of the nominal value.

4.3.4 Test standards. The transmitter shall be tested against an indicating standard selected for maximum error of ±0.1 of a circular degree. A primary transmitting master, such as a dead-weight tester, with a maximum inherent error not exceeding ±1 psi shall be used for testing the transmitter.

* 4.4 First article inspection.

4.4.1 Test samples. The test samples shall consist of twelve transmitters representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. Seven of the samples shall be subjected to the reliability test only, three shall be subjected to all the tests except reliability, and two samples shall be subjected to the scale error and friction tests at room temperature only.

* 4.4.2 First article tests. The first article tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Scale error at room temperature

c. Friction error

d. Position error

e. Dielectric strength

f. Leakage

g. Electrical zero and verification leads.

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* 4.5.2 Sampling plan A tests. Three transmitters shall be selected at random from each 100 or less produced on the contract or order and shall be subjected to the following tests as described under 4.5.1 and 4.6:

a. Individual tests

b. Low temperature

c. High temperature

d. Electrical characteristics

e. Vibration

* 4.5.2.1 Lot acceptance. Each group of 3 samples shall have successfully completed all of the above tests prior to acceptance of the 100 units or portion thereof from which the samples were selected.

* 4.5.3 Sampling plan B tests. Unless otherwise specified, three transmitters shall be selected at random from the first 15 on the contract and subjected to the following tests as described under 4.5.1 and 4.6:

a. Individual tests

b. Sampling plan A tests

c. Low temperature exposure

d. High temperature exposure

e. Temperature-altitude

f. Electromagnetic interference

g. Salt fog

h. Voltage and frequency variation

i. Effect of lead length

j. Case pressure release

k. Burst pressure

l. Damping

m. Seasoning

n. Shock

o. Helium

p. Overpressure

q. Internal examination

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* 4.5.4 Sampling plan C tests. Samples shall be selected in accordance with MIL-STD-105 to yield an acceptable quality level (AQL) of 1.5 percent defective, using the monthly production lot as a basis for sample selection, not to exceed a maximum of 50 units per sample. These units shall be subjected to the following tests as described under 4.5.1 and 4.6:

a. Individual tests

b. Low temperature

c. High temperature

d. Vibration

e. Electrical characteristics.

* 4.5.4.1 Rejection and retest. When one item selected from a production run fails to meet the specification, no items on hand or later produced shall be accepted until the extent and cause of failure have been determined.

* 4.5.5 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

* 4.5.5.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.6 Test methods

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and all other requirements not covered by tests.

4.6.2 Scale error at room temperature. The transmitter shall be tested for scale error at room temperature at the test points specified in table I for both upscale and downscale output readings. The upscale test shall be accomplished by bringing the input pressure up to, but not exceeding, the specified pressure test point. The difference in the output pressure indication after vibration and the input pressure shall not exceed the tolerances specified in table I. The downscale test shall be accomplished in the same manner, except that the input pressure shall be decreased to, but not below, the specified test point.

4.6.3 Friction error. The difference in synchro output prior to vibrating the transmitter and after application of this vibration shall be friction error. Friction error shall not exceed the tolerance specified in table II.

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4.6.4 Position error. This test may be conducted in conjunction with the test specified in 4.6.2. The transmitter shall be energized to obtain a reading of approximately 2,000 psi. The maximum difference in indication caused by the rotation of the transmitter from one position to any other position shall not exceed the tolerance specified in table II for position error. There shall be at least eight test positions spaced 45° apart about each of the three major axes of the transmitter.

4.6.5 Dielectric strength. A potential of 200V rms at a commercial frequency shall be applied between any pin of the electrical connector not connected to the case and the case for a minimum period of 5 seconds. There shall be no insulation breakdown.

4.6.6 Leakage. The transmitter case shall be tested for leaks by means of a mass spectrometer-type helium leak detector. The leak rate shall not exceed 1.0 micron cubic foot per hour at a pressure differential of 12 to 14.7 psi.

4.6.7 Electrical zero and verification of terminal leads. The transmitter shall be wired in accordance with figure 2 and tested for compliance with 3.11.2. Electrical zero shall be attained with a pressure of 2,000 $\pm$5 psi and shall be located at 180° $\pm$0.4°. The signal rotation at the connector pins shall be ADC for increasing values of pressure function. A pressure signal shall be applied to the transmitter to obtain the following conditions which will indicate electrical zero and verify the terminal leads:

a. The voltage between S1 and S3 shall be a minimum, shall be in phase with rotor voltage E (R2R1), and shall remain approximately in phase as the rotor is turned counterclockwise through an angle not exceeding 180° from electrical zero.

b. The voltage between R1 and S3 shall be near the low null and shall reach the low null when the rotor is turned clockwise approximately 30° from electrical zero.

c. Clockwise rotation of the rotor not exceeding 30° from electrical zero shall cause an increase in voltage between S3 and S2 and a decrease in voltage between S2 and S1.

4.6.8 Low temperature. The transmitter shall be subjected to a low *temperature test in accordance with MIL-STD-810, method 502.1, at -54 $\pm$2°C for a period of 4 hours after which it shall be subjected to the scale error and friction error tests while still at that temperature. The errors shall not exceed the tolerances specified in tables I and II, respectively.

4.6.9 High temperature. The transmitter shall be subjected to a high *temperature test in accordance with MIL-STD-810, method 501.1, procedure I, at 145° $\pm$2C for a period of 6 hours. During this test, the transmitter shall be electrically connected to an indicator with an error of not more than 6 minutes of arc, with power applied. At the end of the 6-hour period and while still at the specified temperature, the transmitter shall be subjected to the scale error and friction error tests. The tolerances shall not exceed those specified in tables I and II.

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4.6.10 Vibration. The transmitter shall be subjected to a vibration test * in accordance with MIL-STD-810, Method 514.2, category b.1, procedure I (Part I), table 514.2-II, figure 514.2-2 (curve H), with an input pressure of approximately 2,000 psi. Pointer oscillation and variation shall not exceed the tolerances specified in table II. At the conclusion of this test, the transmitter shall be subjected to the scale error test. The errors shall not exceed the tolerances specified in table I. There shall be no damage to the transmitter as result of this test.

4.6.11 Electrical characteristics. The transmitter shall be tested to determine compliance with 3.4.1 and the requirements specified herein with respect to primary current, power input, secondary voltage, and operation of either one or two indicators, with size 10 synchros, with a change in indication of not more than 1° of arc.

* 4.6.12 Low temperature exposure. The low temperature exposure test shall be conducted in accordance with MIL-STD-810, method 502.1. The transmitter shall be exposed to a temperature of -62° ±2°C for a minimum of 48 hours after which it shall be operated throughout its range at the specified temperature. The temperature shall then be raised to -54° ±2°C and the transmitter maintained at this temperature for a minimum of 24 hours. At the end of the 24-hour period, the transmitter shall be subjected to the low temperature scale error test. The output shall be within the low temperature test tolerances specified in table II. The transmitter shall be returned to room temperature for a minimum period of 4 hours and then subjected to the scale error test at room temperature. The scale error shall not exceed the tolerances specified in table I for room temperature.

* 4.6.13 High temperature exposure. The high temperature exposure test shall be conducted in accordance with MIL-STD-810, Method 501.1. The transmitter shall be exposed to 160° ±5°C for 40 hours. It shall then be returned to room temperature for a minimum of 4 hours after which it shall be subjected to the scale error test at room temperature. The scale error shall not exceed the tolerances specified in table I.

* 4.6.14 Temperature-altitude. The temperature-altitude test shall be conducted in accordance with MIL-STD-810, Method 504.1, equipment category 7, continuous and non-operating modes only. After this test, the transmitter shall be subjected to the scale error test at room temperature. It shall not exceed the tolerances specified in table I.

4.6.15 Electromagnetic interference. The transmitter shall be subjected to and shall meet the electromagnetic interference tests specified in MIL-STD-462, for class ID equipment, up to 400 MHz.

4.6.16 Salt fog. The transmitter shall be subjected to a salt fog test * in accordance with MIL-STD-810, Method 509.1. The transmitter shall be examined and there shall be no evidence of external corrosion or deterioration *that would affect subsequent operations. The external surface may be washed with distilled water and air dried after which the transmitter shall meet the tests specified in 4.6.5 and 4.6.6.

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4.6.17 Voltage and frequency variation. The transmitter shall be subjected to the scale error and friction error tests during which the voltage and frequency shall be varied from 22 to 30V and 320 to 480 Hz (or any combinations thereof). The scale and friction errors shall not exceed those specified in tables I and II, respectively for voltage and frequency variation.

4.6.18 Effect of lead length. The lead length between the transmitter and indicator synchro stators shall be changed from 5 to 300 feet of No. 20 AWG stranded copper wire. The indication shall not change more than 1° of arc.

4.6.19 Overpressure. Sufficient pressure shall be applied gradually to determine compliance with 3.4.1.9. The transmitter shall then be operated at an applied pressure of 5,000 psi for a period of 10 minutes. Not less than 1 hour following this test, the transmitter shall be tested for scale error for increasing pressure only. The scale error shall not exceed the tolerances specified in table I.

4.6.20 Case pressure release. The pressure blowout device shall be tested either under simulated increasing case pressure or by actual injection of pressure into the transmitter case. The device shall operate when the pressure reaches 80 $\pm$25 psi.

4.6.21 Burst pressure. One of the qualification samples shall be subjected to a burst pressure test in accordance with MIL-H-8775 at a temperature of 70°C. The burst test pressure shall be 7,500 psi. The pressure shall then be raised above 7,500 psi to obtain actual burst-pressure data. The burst-test sample delivered to the procuring activity shall be tagged or marked TESTED TO DESTRUCTION - DO NOT INSTALL ON AIRCRAFT.

4.6.22 Damping. A pressure of 4,000 psi shall be applied to the transmitter. The pressure shall be released suddenly and the time required for the pressure to go from 4,000 to 500 psi shall be from 1.0 second to 3.0 seconds. This test shall be conducted with hydraulic fluid conforming to MIL-H-5606 and at a temperature of -55° $\pm$2°C.

4.6.23 Seasoning. The transmitter shall be subjected to 20,000 pressure applications to produce indications from 0 to 85 $\pm$5 percent of full scale and return to 0 at a rate of 3 $\pm$1 cpm. Essentially, the pressure shall vary sinusoidally. The transmitter shall be tested for scale error and friction error at room temperature 1 $\pm$0.1 hour after completion of the cycling. The tolerances specified in tables I and II for room temperature shall apply.

4.6.24 Shock. While operating at approximately 2,000 psi, the transmitter shall be subjected to a shock test in accordance with MIL-STD-810, * Method 516.2, procedure I, figure 516.2-1, for flight vehicle equipment. At the conclusion of this test, the transmitter shall be subjected to the scale error and friction error tests at room temperature. The errors shall not exceed the tolerances specified in tables I and II.

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4.6.25 Helium. The transmitter case shall be punctured, or the filling tube cut, and the case shall then be tested by means of a mass spectrometer-type helium leak detector. Failure to detect helium leakage shall be cause for rejection.

4.6.26 Reliability test program. Prior to starting the reliability test, the contractor shall submit a detailed reliability test program for formal approval by the procuring activity.

4.6.26.1 Reliability test. The samples specified in 4.4.1 shall be subjected to a reliability test. Acceptance permitting production and the total test time shall be in accordance with MIL-STD-781, test plan I. Recording, data handling, and reporting shall also be in accordance with MIL-STD-781. The test equipment shall include provisions for vibration and temperature cycling at the periodic intervals specified herein. During this test, input signals shall be applied to the transmitter at a rate of 3 to 4 cycles per minute from 0 to 85 percent of full scale and return. The test shall begin when rated power is applied. After completion of 4 hours of operation with rated power on, which shall include at least 720 cycles of operation, the power shall be turned off and shall remain off for a period of 10 minutes. For at least 10 minutes out of every hour of ON time, the transmitter shall be vibrated in accordance with MIL-STD-781, test level J. The temperature cycles shall be conducted in the following sequence: 4 hours at -54°C, temperature change to +145°C at a minimum rate of 5°C per minute; 4 hours at +145°C, temperature change to -54°C; and 4 hours at -54°C. In case of malfunction, the malfunction shall be recorded and the testing of that transmitter interrupted while repairs are made. Upon completion of each 250 hours of cycling, the transmitter shall be checked for scale error. The scale errors shall not exceed 150 percent of the room temperature tolerances in table I. No recalibration shall be performed on the units during this test. The test may be interrupted for weekend shutdown. The contractor shall prepare a report in accordance with MIL-STD-831 which shall contain test procedures, details of any malfunctions, and corrective actions taken.

4.6.27 Longevity. After completion of the reliability test, three of the seven reliability test samples shall each continue to be tested for a minimum total of 3,000 hours. The test time accumulated during the reliability test shall be applied to this test. During this test, the transmitter shall continue to demonstrate a minimum acceptable MTBF of 2,000 hours at a 90 percent confidence level.

4.6.28 Internal examination. The case of the transmitter shall be opened *and the internal mechanism examined. Any deterioration or damage which could in any manner prevent the transmitter from meeting operation or maintenance requirements during service life, shall be cause for ejection.

4.7 Inspection of preservation, packaging, and packing. The inspection of preservation, packaging, and packing shall be in accordance with section 5.

* 5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A, C or Commercial/Industrial as specified (see 6.2).

* 5.1.1 Level A. Transmitter shall be preserved and packaged in accordance with Method III of MIL-P-116.

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* 5.1.2 Level C. The Level C preservation for Transmitter shall conform to the MIL-STD-794 requirements of this level. OR

Commercial/industrial preservation of transmitter shall be in accordance with the requirements of ASTM D 3951 -82.

* 5.2 Packing. Packing shall be level A, B or C or Commercial/Industrial, as specified (see 6.2).

* 5.2.1 Level A. Transmitters preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-601, overseas type. The closure shall be in accordance with the appendix of the container specification.

* 5.2.2 Level B. Transmitter preserved and packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-636, Weather Resistant. The closure shall be in accordance with the appendix of the container specification.

* 5.2.3 Level C. Transmitter preserved and packaged as specified in 5.1.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Container shall be in accordance with either the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation. OR

Commercial/industrial packing of transmitter shall be in accordance with the requirements of ASTM D 3951-82.

* 5.3 Physical Protection. Cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning materials and container shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

* 5.4 Marking. In addition to any other marking required by the order or contract, interior and exterior containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be: Transmitter, Pressure, Hydraulic, 0-4000 PSI, Type MJ-1.

5.4.1 Reinspection date. The reinspection date marking shall be as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The Type MJ-1 transmitter covered by this specification is intended for use in aircraft hydraulic systems, in which the nominal system pressure is not rated above 3,000 psi, to transmit the hydraulic pressure in the hydraulic system of the aircraft to an applicable indicator. The transmitter is also intended for use in locations wherein the temperature does not exceed 145°C during transmitter operation and does not exceed 160°C when it is not operating.

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* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Information, if any, to be submitted with the qualification samples

c. Levels of packaging and packing required and reinspection date marking (see section 5).

* d. Option of either 4.5.2 or 4.5.4 applied to each procurement or

* e. If sampling plan B test is not to be conducted.

* 6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample as it may be a standard production item from the contractor's current inventory. The first article should consist of twelve units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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directed by the contracting officer.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-25307D	Transmitter, Pressure, Hydraulic, Type MJ-1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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